1. Let $f(z) = \sum_{n=0}^{\infty} a_n z^n$ and $g(z) = \sum_{n=0}^{\infty} b_n z^n$ be two analytic functions on the unit disk $\mathbb{D} := \mathbb{D}(0;1)$. For any $n \in \mathbb{N}$, set

$$c_n = \sum_{k=0}^{n} a_k b_{n-k}.$$

Show that $f(z)g(z)$ is analytic on $\mathbb{D}$, with the Taylor expansion $f(z)g(z) = \sum_{n=0}^{\infty} c_n z^n$.
Hint. Show that $c_n = (fg)^{(n)}(0)/n!$, using the *Leibniz's rule*:

$$(fg)^{(n)}(z) = \sum_{k=0}^{n} C_n^k f^{(k)}(z) g^{(n-k)}(z).$$

2. The *Legendre polynomials* can be defined as the coefficients $P_n(\alpha)$ of the series expansion of

$$f(z) = \frac{1}{(1 - 2\alpha z + z^2)^{1/2}} = 1 + \sum_{n=1}^{\infty} P_n(\alpha) z^n.$$

Calculate the first four Legendre polynomials.

3. Let $\{a_n\}_{n=0}^{\infty}$ and $\{b_n\}_{n=0}^{\infty}$ be two sequences of complex numbers, and set

$$c_n = \sum_{k=0}^{n} a_k b_{n-k}, \quad n \in \mathbb{N}.$$

Suppose that $\sum_{n=0}^{\infty} a_n$, $\sum_{n=0}^{\infty} b_n$ and $\sum_{n=0}^{\infty} c_n$ are all convergent. Show that

$$\left(\sum_{n=0}^{\infty} a_n \right) \cdot \left(\sum_{n=0}^{\infty} b_n \right) = \sum_{n=0}^{\infty} c_n.$$

Hint. Use Exercise 1 and Abel's theorem.

4. Let $\{a_n\}_{n=0}^{\infty}$ be the *Fibonacci sequence*, i.e. $a_0 = a_1 = 1$ and $a_n = a_{n-1} + a_{n-2}$, $n \geq 2$.
1) Show that the power series $\sum_{n=0}^{\infty} a_n z^n$ coincides with the rational function $1/(1 - z - z^2)$, and determine the radius of convergence.
2) Deduce the following formula of Binet:

$$a_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1 + \sqrt{5}}{2} \right)^{n+1} - \left(\frac{1 - \sqrt{5}}{2} \right)^{n+1} \right].$$

5. Consider the Taylor series of $1/\cos z$ centered at 0:

$$\frac{1}{\cos z} = \sum_{n=0}^{\infty} \frac{E_{2n}}{(2n)!} z^{2n},$$

where E_{2n} are called *Euler numbers*.
1) Determine the radius of convergence of the series.
2) Show that all E_{2n} are natural numbers, and compute E_{2n} for $0 \leq n \leq 6$.